COLORADO · THE NEXT ERA OF COMPUTE

Silicon Valley > Photon Mountain

Photon Mountain

Silicon Valley was named for the material of the last era. Photon Mountain is named for the medium of the next, light. Along Colorado's Front Range, one corridor is building the AI age out of photons: light-based chips, renewable energy, and quantum systems.

WHAT IS PHOTON MOUNTAIN

Photon Mountain is a real place, the Front Range corridor where Colorado builds the AI age out of light. A photon is the particle of light, so the name holds both the photonic chips Colorado can make and the quantum science it can run. Two rings: a dense core of labs and fabs, and a greater ring of cities.

CORE The dense Boulder, Golden, Arvada, and Denver cluster: the labs, the fabs, and the metro hub.



GREATER The Front Range ring around the core, where the corridor's companies and workforce spread out.

Lafayette · Louisville · Broomfield · Westminster · Longmont

PHOTONIC CHIPS

Chips that compute and move data with light, at far lower power. Colorado fabricates them today at NIST Boulder.

>$10B 2034 market, per analyst forecasts

RENEWABLE ENERGY

Power is the constraint on the AI build. Golden's National Laboratory of the Rockies (formerly NREL) is a Department of Energy national lab.

100% renewable-electricity goal by 2040 (Polis roadmap)

QUANTUM SYSTEMS

The Elevate Quantum hub, and Mines' first-in-nation quantum systems engineering bachelor's.

$127M+ federal and state awards, 2024

SILICON VALLEY	PHOTON MOUNTAIN
The computer age	**The AI age**
Silicon, the material	**Photon, the light**
San Jose to Menlo Park	**Denver to Boulder**
Microchips	**Photonic chips, renewable energy, quantum**

Silicon Valley built the last era out of silicon. Photon Mountain builds the next out of light.

Build with light.

Photon Mountain · a Colorado Startups project in collaboration with Colorado AI

Sources: NIST; Elevate Quantum / Colorado OEDIT; Colorado School of Mines; National Laboratory of the Rockies (formerly NREL); Colorado Energy Office; Xcel; IMARC, Precedence and other analysts. NIST Boulder runs an operating photonics cleanroom; the Arvada fab is targeted for 2026, CU's nanofab for 2028. The 100% renewable-electricity target is a 2040 goal.